Supplementary Material for
EMMS: Evidential Multi-Label Multi-Dimensional Selection

This PDF file contains eight sections:

(1) Theoretical proof of the convergence of the proposed EMMS algorithm.

(2) Three tables comparing the performance of different methods in terms of MaF, MiF, and HL.

(3) Five figures illustrating the performance of various methods under different feature selection ratios across five datasets.

(4) Five figures illustrating the performance of various methods under different instance selection ratios across five datasets.

(5) Five figures illustrating the performance of various methods under different label selection ratios across five datasets.

(6) One figure showing the convergence of the proposed EMMS on three additional datasets.

(7) One table and one figure present the statistical analysis results of the Friedman test and the Nemenyi test.

(8) One table comparing the computational time of EMMS and competing methods.

1 Theoretical proof of the convergence of the proposed EMMS algorithm

Since the objective function in Eq. (3) is not jointly convex with respect to $\boldsymbol{P}$, $\boldsymbol{Q}$, $\boldsymbol{R}$, $\boldsymbol{N}$, and $\boldsymbol{Y}$, we solve it by alternately optimizing five subproblems, each with a multiplicative update rule, as summarized in Algorithm 1. To establish the convergence of Algorithm 1, we first introduce the following definition and lemma from [Liu and Wu, 2010].

Definition 1. [Liu and Wu, 2010] A function $\mathcal{G}(x, x')$ is called an auxiliary function for $\mathcal{F}(x)$ if

$$\mathcal{G}(x, x') \geq \mathcal{F}(x), \mathcal{G}(x, x) = \mathcal{F}(x). \tag{1}$$

Lemma 1. [Liu and Wu, 2010] If $\mathcal{G}$ is an auxiliary function for $\mathcal{F}$, then $\mathcal{F}$ is nonincreasing under the following update rule:

$$x^{t+1} = \arg\min_{x} \mathcal{G}(x, x^t), \tag{2}$$

where t denotes the t-th iteration.

Since all variables are updated similarly, we present only the monotonicity proof for $\boldsymbol{P}$, as the others follow analogously. With the other variables fixed, the objective function w.r.t. $\boldsymbol{P}$ is given by $\mathcal{F}(\boldsymbol{P}) = \frac{1}{2}\|\boldsymbol{X}\boldsymbol{P}\boldsymbol{Q}^\top - \boldsymbol{R} - \boldsymbol{Y}\|_F^2 + \lambda_1 \operatorname{Tr}(\boldsymbol{1}_{d\times c}^\top \boldsymbol{P}) + \frac{1}{2}\operatorname{Tr}(\boldsymbol{P}^\top \boldsymbol{L}_F \boldsymbol{P})$. We then have the following lemma.

Lemma 2. Let $\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}') = \mathcal{F}(\boldsymbol{P}') + \operatorname{Tr}((\boldsymbol{P} - \boldsymbol{P}')^\top \nabla \mathcal{F}(\boldsymbol{P}')) + \frac{1}{2}\sum_{i,j} \frac{(\boldsymbol{X}\boldsymbol{P}'\boldsymbol{Q}^\top\boldsymbol{Q} + \boldsymbol{D}_F\boldsymbol{P}')_{ij}}{p'_{ij}}(p_{ij} - p'_{ij})^2$, where $\nabla\mathcal{F}(\boldsymbol{P}')$ denotes the gradient of $\mathcal{F}$ at $\boldsymbol{P}'$. Then $\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}')$ is an auxiliary function for $\mathcal{F}(\boldsymbol{P})$. Moreover, $\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}')$ is convex with respect to $\boldsymbol{P}$, and its global minimum is attained by the update rule in Eq. (9).

Proof. Since $\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}) = \mathcal{F}(\boldsymbol{P})$ holds trivially, it remains to show that $\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}') \geq \mathcal{F}(\boldsymbol{P})$. To this end, we consider the Taylor expansion of $\mathcal{F}(\boldsymbol{P})$ at $\boldsymbol{P}'$, which is given by $\mathcal{F}(\boldsymbol{P}) = \mathcal{F}(\boldsymbol{P}') + \operatorname{Tr}((\boldsymbol{P} - \boldsymbol{P}')^\top \nabla\mathcal{F}(\boldsymbol{P}')) + \frac{1}{2}\sum_{i,j}\sum_{k,l} \frac{\partial^2 \mathcal{F}(\boldsymbol{P}')}{\partial p_{ij}\partial p_{kl}}(p_{ij} - p'_{ij})(p_{kl} - p'_{kl})$. Here, $\frac{\partial^2 \mathcal{F}(\boldsymbol{P}')}{\partial p_{ij}\partial p_{kl}} = (\boldsymbol{X}^\top\boldsymbol{X})_{ik}(\boldsymbol{Q}^\top\boldsymbol{Q})_{jl} + (\boldsymbol{L}_F)_{ik}\delta_{jl}$, where δ_{jl} denotes the Kronecker delta, i.e., $\delta_{jl} = 1$ if $j = l$, and $\delta_{jl} = 0$ otherwise.

Using the inequality $ab \leq \frac{1}{2}\left(\frac{a^2 x}{y} + \frac{b^2 y}{x}\right)$ for $x, y > 0$ [Hardy *et al.*, 1952], we obtain

$$\begin{aligned}&(p_{ij} - p'_{ij})(p_{kl} - p'_{kl}) \\ &\leq \frac{1}{2}\left(\frac{p'_{kl}}{p'_{ij}}(p_{ij} - p'_{ij})^2 + \frac{p'_{ij}}{p'_{kl}}(p_{kl} - p'_{kl})^2\right).\end{aligned} \tag{3}$$

It then follows that

$$\begin{aligned}&\sum_{i,j}\sum_{k,l}(\boldsymbol{X}^\top\boldsymbol{X})_{ik}(\boldsymbol{Q}^\top\boldsymbol{Q})_{jl}(p_{ij} - p'_{ij})(p_{kl} - p'_{kl}) \\ &\leq \sum_{i,j}\frac{(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{P}'\boldsymbol{Q}^\top\boldsymbol{Q})_{ij}}{p'_{ij}}(p_{ij} - p'_{ij})^2.\end{aligned} \tag{4}$$

Similarly, since $\boldsymbol{L}_F = \boldsymbol{D}_F - \boldsymbol{S}_F$ with $\boldsymbol{S}_F \geq 0$, we have

$$\begin{aligned}&\sum_{i,j}\sum_{k,l}(\boldsymbol{L}_F)_{ik}\delta_{jl}(p_{ij} - p'_{ij})(p_{kl} - p'_{kl}) \\ &\leq \sum_{i,j}\frac{(\boldsymbol{D}_F\boldsymbol{P}')_{ij}}{p'_{ij}}(p_{ij} - p'_{ij})^2.\end{aligned} \tag{5}$$

Combining inequalities (4) and (5), we obtain $\mathcal{F}(\boldsymbol{P}) \leq \mathcal{G}(\boldsymbol{P}, \boldsymbol{P}')$. Hence, $\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}')$ is an auxiliary function for $\mathcal{F}(\boldsymbol{P})$ according to Definition 1.

Next, we show that the update rule in Eq. (9) minimizes $\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}^{(t)})$. Since each second-order partial derivative of

$\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}^{(t)})$ with respect to p_{ij} is nonnegative, $\mathcal{G}(\boldsymbol{P}, \boldsymbol{P}^{(t)})$ is convex with respect to $\boldsymbol{P}$. Therefore, its global minimum is attained by setting the first-order derivative with respect to p_{ij} to zero, which leads to the update rule in Eq. (9). Hence, $\boldsymbol{P}^{(t+1)} = \arg\min_{\boldsymbol{P}} \mathcal{G}(\boldsymbol{P}, \boldsymbol{P}^{(t)})$.

According to Lemma 1, we have $\mathcal{F}(\boldsymbol{P}^{(t+1)}) \leq \mathcal{G}(\boldsymbol{P}^{(t+1)}, \boldsymbol{P}^{(t)}) \leq \mathcal{G}(\boldsymbol{P}^{(t)}, \boldsymbol{P}^{(t)}) = \mathcal{F}(\boldsymbol{P}^{(t)})$. Thus, the objective value in Eq. (8) decreases monotonically under Algorithm 1.

2 Three tables comparing the performance of different methods in terms of MaF, MiF, and HL

Methods	Staexp	Corel5k	Arts	Science	Social	Co16k1
EMMS	**0.0404**	**0.0331**	**0.1887**	**0.1675**	**0.4281**	**0.0493**
mFILS	0.0012	0.0009	0.0175	0.0020	0.0127	0.0052
s2LS	0.0001	0.0035	0.0131	0.0915	0.2511	0.0229
MALS	0.0022	0.0012	0.0337	0.0266	0.2285	0.0098
MJLS	0.0032	0.0035	0.0311	0.0369	0.3217	0.0037
IFLS	0.0083	0.0030	0.0949	0.0802	0.2952	0.0064
FILS	0.0084	0.0025	0.1362	0.0658	0.3675	0.0092
RILS	0.0022	0.0027	0.0409	0.0378	0.2795	0.0061
SILS	0.0034	0.0023	0.0432	0.0371	0.2932	0.0051
NILS	0.0024	0.0023	0.0221	0.0435	0.3190	0.0064
PILS	0.0029	0.0008	0.0312	0.0210	0.3015	0.0011

Table 1: Performance comparison between EMMS and competing methods on six datasets in terms of MaF. Larger values indicate better performance.

Methods	Staexp	Corel5k	Arts	Science	Social	Co16k1
EMMS	**0.1622**	**0.1416**	**0.2512**	**0.3075**	**0.5323**	**0.1803**
mFILS	0.0123	0.0089	0.0333	0.0069	0.0365	0.0069
s2LS	0.0008	0.0076	0.0159	0.0855	0.3707	0.0389
MALS	0.0102	0.0045	0.0391	0.0240	0.3716	0.0168
MJLS	0.0172	0.0065	0.0372	0.0534	0.4771	0.0048
IFLS	0.0228	0.0150	0.1027	0.0795	0.4205	0.0075
FILS	0.0083	0.0104	0.1437	0.0693	0.5004	0.0384
RILS	0.0096	0.0058	0.0402	0.0456	0.3873	0.0139
SILS	0.0124	0.0062	0.0463	0.0313	0.4496	0.0092
NILS	0.0114	0.0039	0.0224	0.0521	0.4838	0.0103
PILS	0.0024	0.0018	0.0376	0.0266	0.4323	0.0012

Table 2: Performance comparison between EMMS and competing methods on six datasets in terms of MiF. Larger values indicate better performance.

Methods	Staexp	Corel5k	Arts	Science	Social	Co16k1
EMMS	**0.0020**	**0.0009**	**0.0072**	**0.0049**	**0.0037**	**0.0039**
mFILS	0.0115	0.0082	0.0482	0.0195	0.0274	0.0177
s2LS	0.0064	0.0409	0.2103	0.1390	0.2050	0.0867
MALS	0.0066	0.0405	0.1989	0.1492	0.2132	0.0950
MJLS	0.0065	0.0402	0.1788	0.1466	0.1873	0.0945
IFLS	0.0245	0.0417	0.1774	0.1370	0.1895	0.0934
FILS	0.0157	0.0584	0.1652	0.1424	0.1694	0.1064
RILS	0.0198	0.0461	0.2099	0.1585	0.2059	0.0951
SILS	0.0065	0.0404	0.1936	0.1481	0.1805	0.0953
NILS	0.0065	0.0403	0.2043	0.1507	0.1823	0.0953
PILS	0.0065	0.0403	0.2014	0.1505	0.1866	0.0944

Table 3: Performance comparison between EMMS and competing methods on six datasets in terms of HL. Smaller values indicate better performance.

3 Five figures illustrating the performance of various methods under different feature selection ratios across five datasets

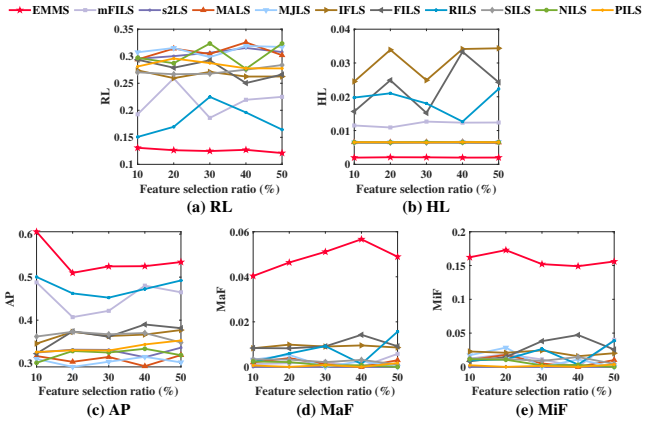

Figure 1: Comparison of different methods on Staexp dataset across varying feature selection ratios using five evaluation metrics.

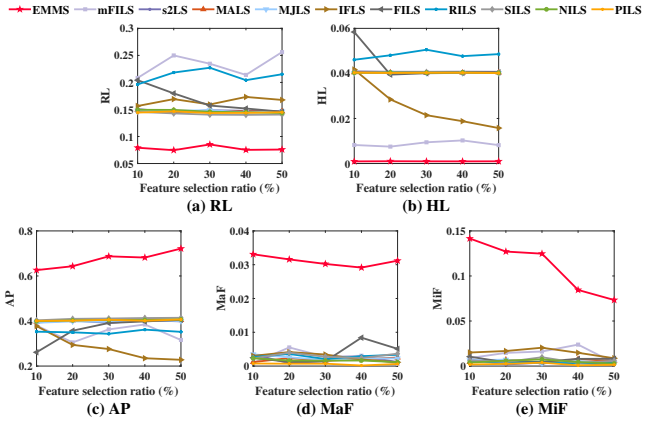

Figure 2: Comparison of different methods on Corel5k dataset across varying feature selection ratios using five evaluation metrics.

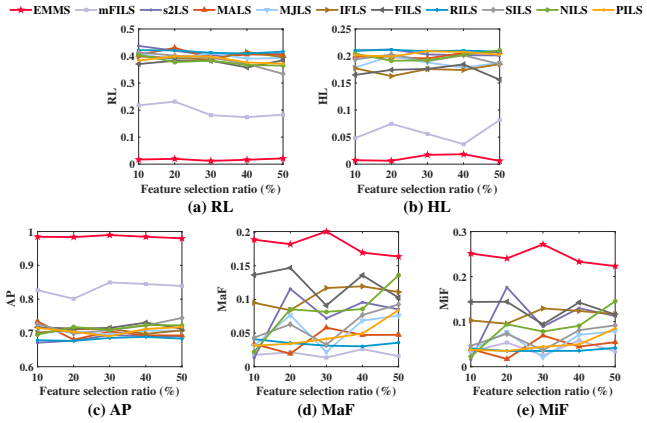

Figure 3: Comparison of different methods on Arts dataset across varying feature selection ratios using five evaluation metrics.

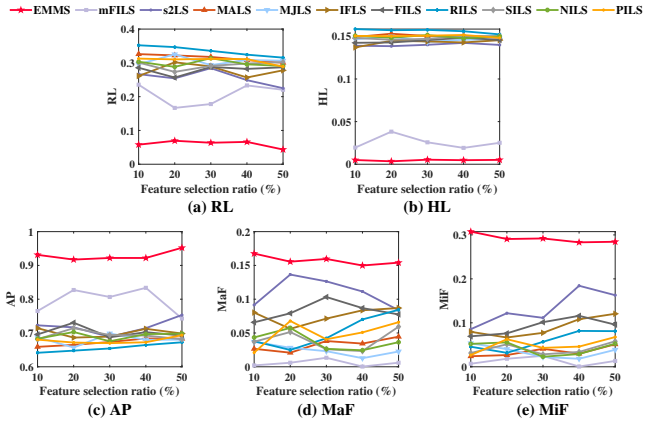

Figure 4: Comparison of different methods on Science dataset across varying feature selection ratios using five evaluation metrics.

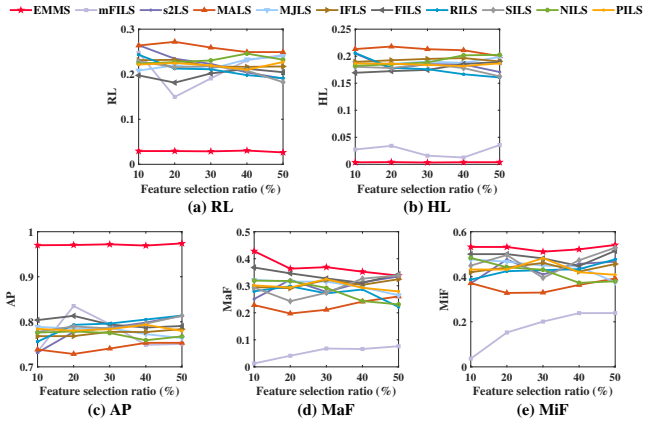

Figure 5: Comparison of different methods on Social dataset across varying feature selection ratios using five evaluation metrics.

4 Five figures illustrating the performance of various methods under different instance selection ratios across five datasets

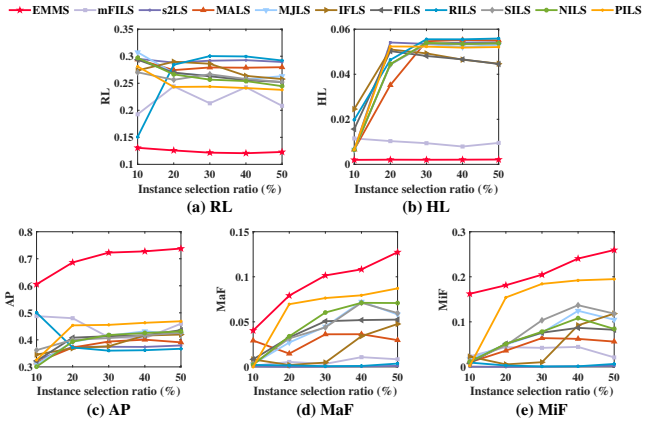

Figure 6: Comparison of different methods on Staexp dataset across varying instance selection ratios using five evaluation metrics.

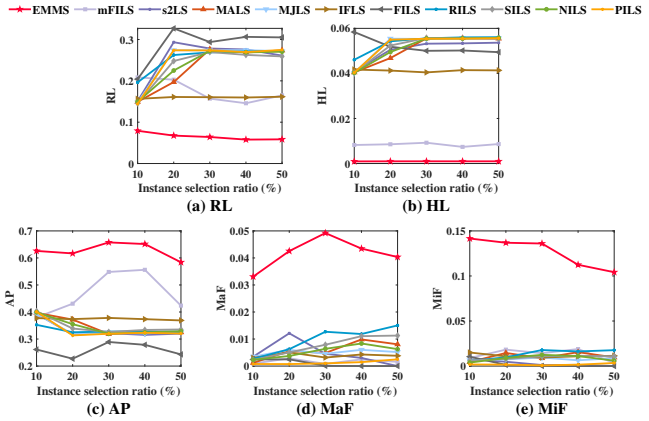

Figure 7: Comparison of different methods on Corel5k dataset across varying instance selection ratios using five evaluation metrics.

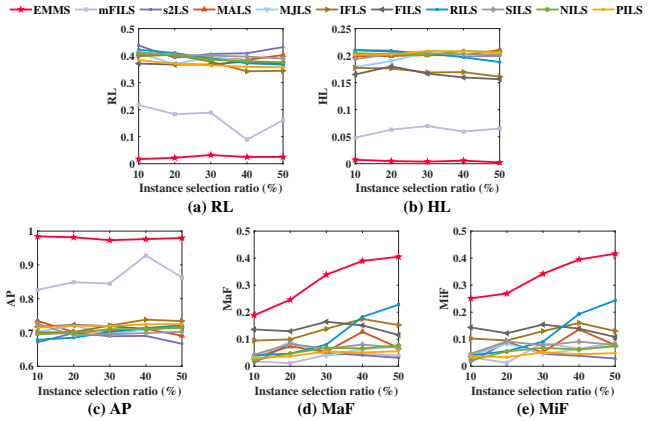

Figure 8: Comparison of different methods on Arts dataset across varying instance selection ratios using five evaluation metrics.

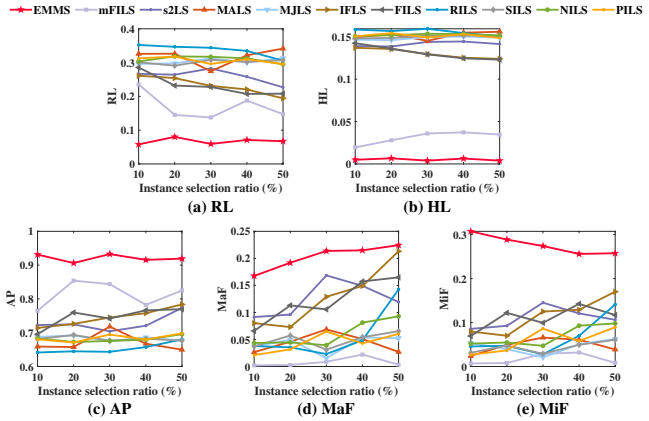

Figure 9: Comparison of different methods on Science dataset across varying instance selection ratios using five evaluation metrics.

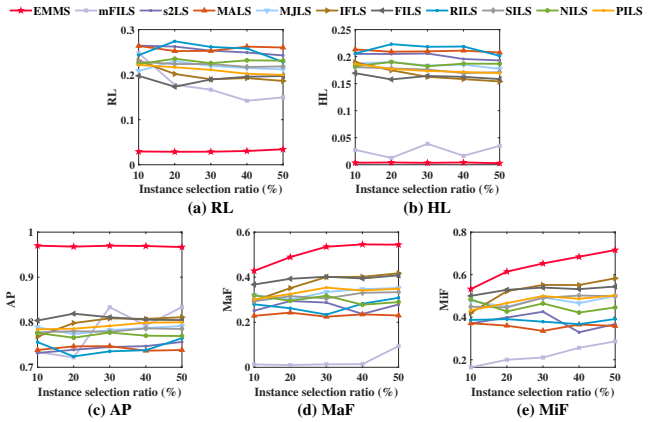

Figure 10: Comparison of different methods on Social dataset across varying instance selection ratios using five evaluation metrics.

5 Five figures illustrating the performance of various methods under different label selection ratios across five datasets

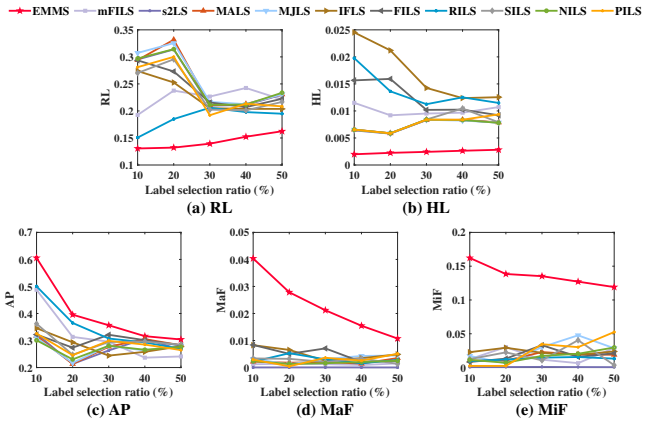

Figure 11: Comparison of different methods on Staexp dataset across varying label selection ratios using five evaluation metrics.

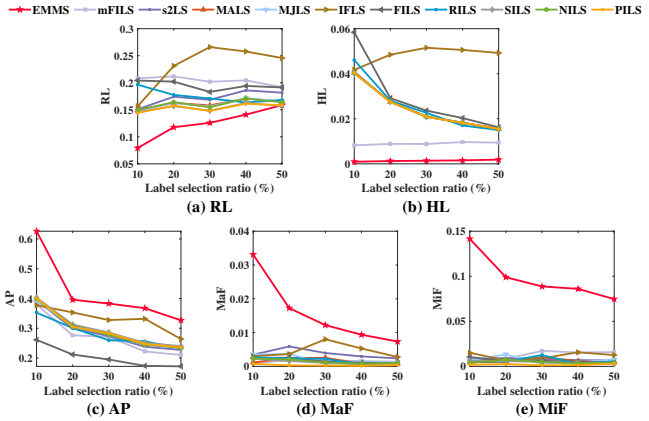

Figure 12: Comparison of different methods on Corel5k dataset across varying label selection ratios using five evaluation metrics.

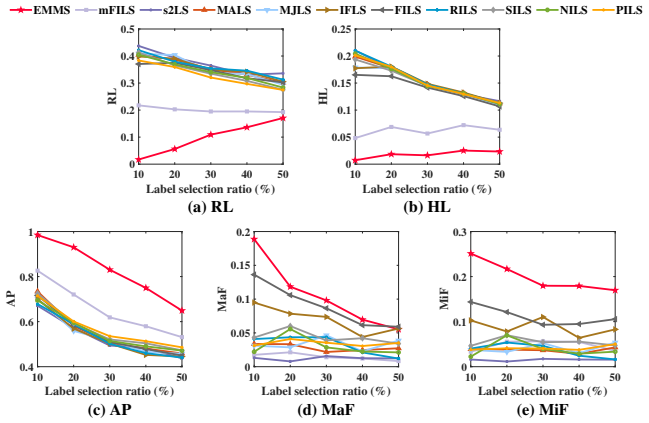

Figure 13: Comparison of different methods on Arts dataset across varying label selection ratios using five evaluation metrics.

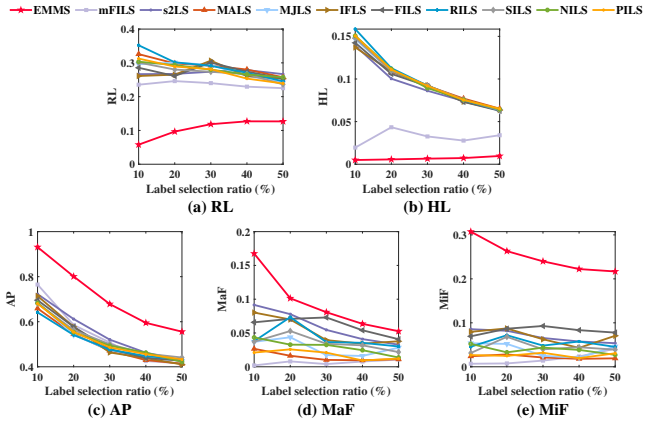

Figure 14: Comparison of different methods on Science dataset across varying label selection ratios using five evaluation metrics.

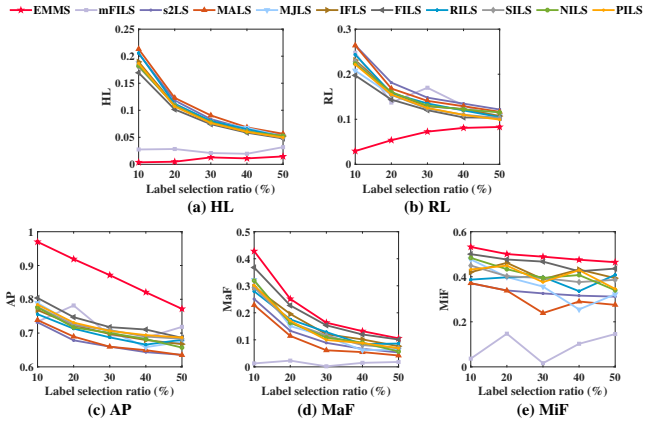

Figure 15: Comparison of different methods on Social dataset across varying label selection ratios using five evaluation metrics.

6 One figure showing the convergence of the proposed EMMS on three additional datasets

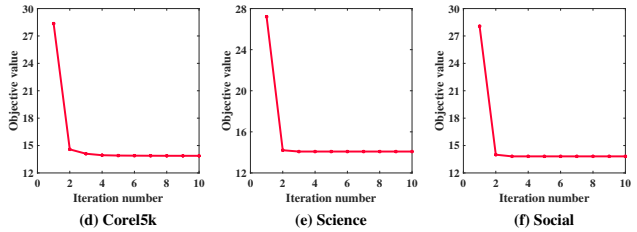

Figure 16: Convergence curves of EMMS on Corel5k, Science, and Social datasets.

7 One table and one figure present the statistical analysis results of the Friedman test and the Nemenyi test

Metrics	F_F	Critical value ($\alpha = 0.05$)
AP	19.3030	
MaF	32.7273	
MiF	27.8788	18.3070
HL	30.3636	
RL	20.0000	

Table 4: Friedman statistic F_F and critical value across five evaluation metrics.

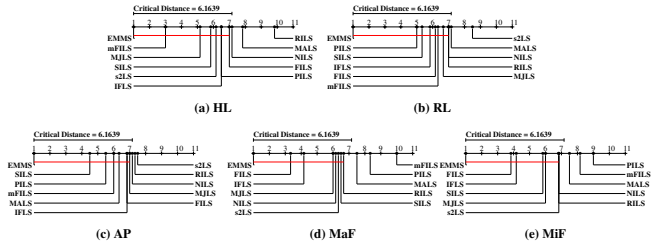

Figure 17: Comparison of EMMS with other methods using the Nemenyi test.

8 One table comparing the computational time of EMMS and competing methods

The computational complexity of EMMS is analyzed in the Complexity and Convergence Analysis subsection of the main manuscript. To further evaluate its efficiency empirically, we compare the running times of all methods. Table 5 reports the average running times over five runs on each dataset. EMMS achieves shorter running times than the competing methods, and the improvements are statistically significant according to the Wilcoxon signed-rank test.

Methods	Staexp	Corel5k	Arts	Science	Social	Co16k1
EMMS	3.69*	10.23*	1.80*	2.55*	6.63*	31.66*
mFILS	94.72	195.38	24.06	37.57	32.32	359.98
s2LS	12.35	15.66	15.06	16.03	19.53	116.50
MALS	48.76	81.93	29.97	34.51	41.32	282.29
MJLS	2589.76	3027.34	265.24	563.86	750.84	4180.92
IFLS	126.32	172.28	46.63	38.41	35.68	537.17
FILS	146.28	187.52	51.28	55.92	46.94	551.66
RILS	383.05	149.87	96.45	184.17	336.96	1859.04
SILS	278.86	126.78	103.45	175.67	285.52	1442.42
NILS	238.25	150.70	113.76	167.45	297.44	1409.72
PILS	286.08	125.54	100.59	175.56	288.86	1639.34

Table 5: Average running time (s) of different methods (* indicates statistical significance by Wilcoxon signed-rank test).

References

[Hardy *et al.*, 1952] Godfrey Harold Hardy, John Edensor Littlewood, and George Pólya. *Inequalities*. Cambridge University Press, 1952.

[Liu and Wu, 2010] Haifeng Liu and Zhaohui Wu. Non-negative matrix factorization with constraints. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 24, pages 506–511, 2010.